

Microservices with Spring Boot 3 and Spring Cloud

2. Microservices with API gateway

Creating Microservices for account and loan

A screenshot of a web browser window. The address bar shows 'localhost:8080/accounts/00987987973...'. The page content displays a JSON object:

```
{
  "number": "00987987973432",
  "type": "savings",
  "balance": 234343
}
```

A screenshot of the Eclipse IDE console window. It shows the output of a Spring Boot application starting. The logs include:

```

:: Spring Boot ::
(v4.1.0)

2026-07-12T16:45:14.499+05:30 INFO 10944 --- [account] [ restartedMain ] Jar.AccountApplication : Starting AccountApplication using Java 17.0.18 with PID 10944 (D:\Cogniz_folder\Solut
2026-07-12T16:45:14.507+05:30 INFO 10944 --- [account] [ restartedMain ] Jar.AccountApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-12T16:45:14.640+05:30 INFO 10944 --- [account] [ restartedMain ] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! Set 'spring.devtools.add-properties' to 'false' to
2026-07-12T16:45:14.641+05:30 INFO 10944 --- [account] [ restartedMain ] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property
2026-07-12T16:45:16.012+05:30 INFO 10944 --- [account] [ restartedMain ] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-12T16:45:16.026+05:30 INFO 10944 --- [account] [ restartedMain ] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-12T16:45:16.026+05:30 INFO 10944 --- [account] [ restartedMain ] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-12T16:45:16.072+05:30 INFO 10944 --- [account] [ restartedMain ] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1429 ms
2026-07-12T16:45:16.505+05:30 INFO 10944 --- [account] [ restartedMain ] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-12T16:45:16.515+05:30 INFO 10944 --- [account] [ restartedMain ] Jar.AccountApplication : Started AccountApplication in 2.561 seconds (process running for 3.064)
2026-07-12T16:46:44.880+05:30 INFO 10944 --- [nio-8080-exec-2] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-12T16:46:44.881+05:30 INFO 10944 --- [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-12T16:46:44.884+05:30 INFO 10944 --- [nio-8080-exec-2] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
```

A screenshot of the Eclipse IDE. The Package Explorer on the left shows the project structure:

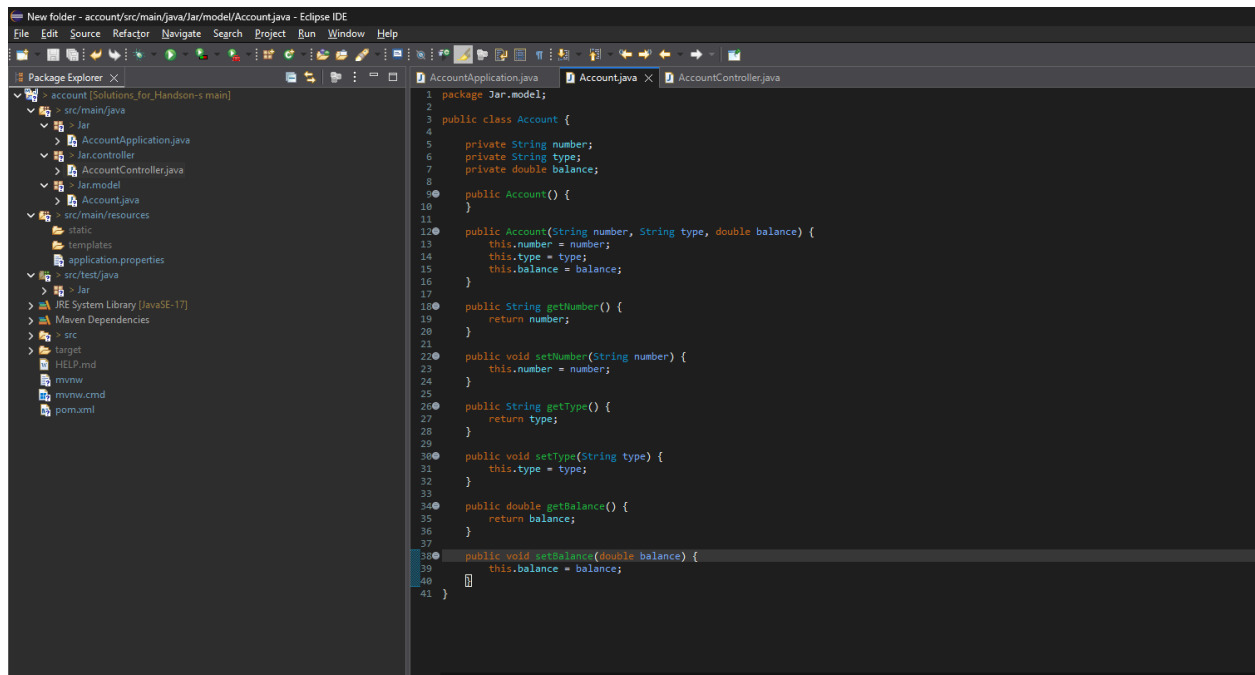
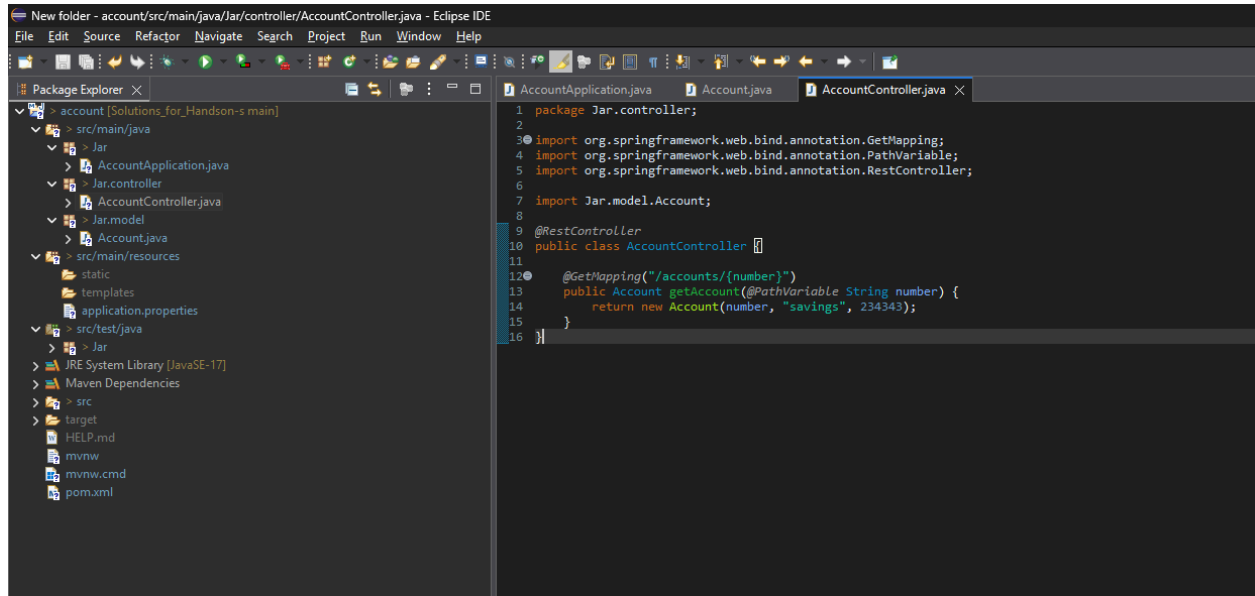
```

account [Solutions_for_Handson-s main]
├── src/main/java
│   ├── Jar
│   │   ├── AccountApplication.java
│   │   ├── Jar.controller
│   │   ├── AccountController.java
│   │   ├── Jar.model
│   │   └── Account.java
│   └── src/main/resources
│       ├── static
│       ├── templates
│       └── application.properties
├── src/test/java
│   └── Jar
├── JRE System Library [JavaSE-17]
├── Maven Dependencies
├── src
├── target
├── HELP.md
├── mvnw
├── mvnw.cmd
└── pom.xml
```

The main editor shows the `AccountApplication.java` file with the following code:

```

1 package Jar;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class AccountApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(AccountApplication.class, args);
11     }
12
13 }
14
```



Part 2

```
localhost:8081/loans/H00987987972342

Pretty-print ✓

{
  "number": "H00987987972342",
  "type": "car",
  "loan": 400000,
  "emi": 3258,
  "tenure": 18
}
```

```
Problems Javadoc Declaration Console X
LoanApplication [Java Application] C:\Program Files\Eclipse Adoptium\jdk-17.0.18-hotspot\bin\javaw.exe (12-Jul-2026, 5:12:50 pm elapsed 0:00:48) [pid: 8944]

:: Spring Boot ::
(v4.1.0)

2026-07-12T17:12:52.579+05:30 INFO 8944 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Starting LoanApplication using Java 17.0.18 with PID 8944 (D:\Cogniz_folder\Solutions_for_
2026-07-12T17:12:52.589+05:30 INFO 8944 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-12T17:12:52.762+05:30 INFO 8944 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties' to 'false' to dis
2026-07-12T17:12:52.763+05:30 INFO 8944 --- [loan] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'I
2026-07-12T17:12:54.513+05:30 INFO 8944 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8081 (http)
2026-07-12T17:12:54.540+05:30 INFO 8944 --- [loan] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-12T17:12:54.540+05:30 INFO 8944 --- [loan] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-12T17:12:54.630+05:30 INFO 8944 --- [loan] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1864 ms
2026-07-12T17:12:55.404+05:30 INFO 8944 --- [loan] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8081 (http) with context path '/'
2026-07-12T17:12:55.419+05:30 INFO 8944 --- [loan] [ restartedMain] com.cognizant.loan.LoanApplication : Started LoanApplication in 3.85 seconds (process running for 4.682)
2026-07-12T17:13:08.953+05:30 INFO 8944 --- [loan] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-12T17:13:08.954+05:30 INFO 8944 --- [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-12T17:13:08.956+05:30 INFO 8944 --- [loan] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
```

```
New folder - loan/src/main/java/com/cognizant/loan/LoanApplication.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer X
> account [Solutions_for_Handson-s main]
  > loan [Solutions_for_Handson-s main]
    > src/main/java
      > com.cognizant.loan
        > LoanApplication.java
      > com.cognizant.loan.controller
        > LoanController.java
      > com.cognizant.loan.model
        > Loan.java
    > src/main/resources
      static
      templates
      application.properties
  > src/test/java
  > JRE System Library [JavaSE-17]
  > Maven Dependencies
  > src
  > target
    HELP.md
    mvnw
    mvnw.cmd
    pom.xml

AccountApplication... Account.java AccountControllerj... Loan.java
1 package com.cognizant.loan;
2
3 import org.springframework.boot.SpringApplication;
4
5 @SpringBootApplication
6 public class LoanApplication {
7
8     public static void main(String[] args) {
9         SpringApplication.run(LoanApplication.class, args);
10     }
11 }
12
13
14
```

